

Counterfactual curation

First local execution

Measured engineering progress | 4 October 2026 | All computation in this workspace

Implemented and exercised the core repair pipeline on natural text. The independent curation oracle, eligible-payload baseline, indexed coefficient summary, source/record structural analysis and cumulative request harness are now runnable. No remote compute jobs were used.

Scope: these are implementation checks, not the ACL paper's confirmatory experiments. The data are small, connector-rendered convenience previews. Features are fixed lexical hashing, not E5 or MPNet; thresholds lack human semantic calibration. No synthetic documents or labels were introduced.

Verified results

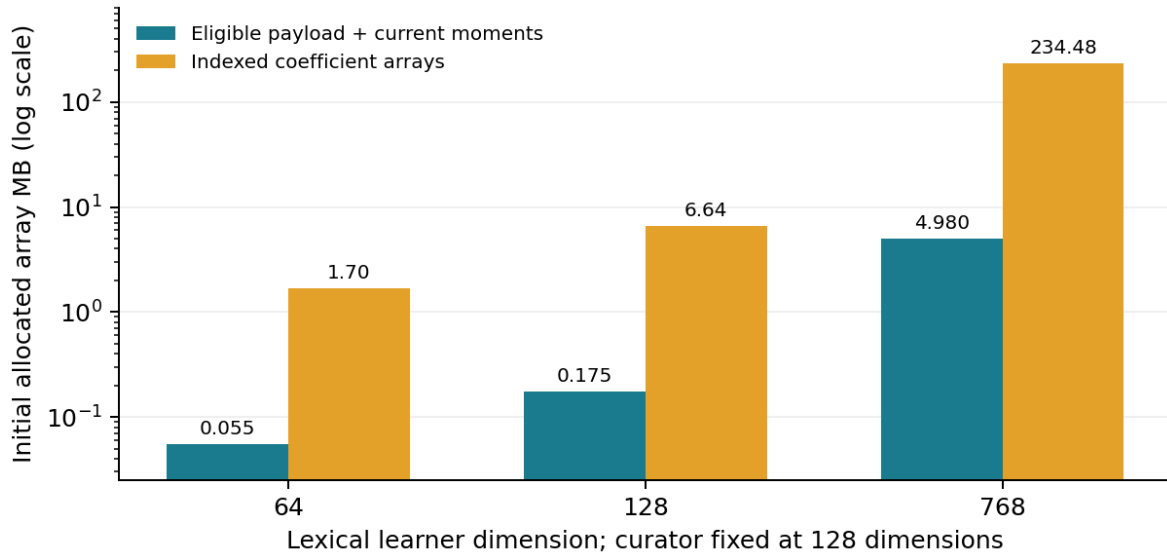
Check	Measured outcome	Interpretation
Natural records	100 Civil Comments + 90 CC-News articles	Civil retains original toxicity labels. News receives no invented target.
Joint method/oracle checks	324 release checkpoints; 0 failed trajectories	Selection, count, moments and heads checked; not independent population trials.
Numerical agreement	Max head difference 6.66e-16; max moment difference 8.88e-16	FP64 diagnostic agreement, not a rigorous roundoff certificate.
Complete coefficient rebuild	8 preselected checkpoints at learner dimensions 64 and 128	All keys/coordinates compared; this audit does not cover every run.
Full deletion	100 additional releases; zero final head	Raw cancellation residue remains visible rather than hidden.
News source selection	96 subset checks; no mismatches	All subsets of five observed hosts at three lexical thresholds.
Fresh-process replay	4 outputs match; zero observed guarded Python I/O attempts	Receives state and request IDs only; not an OS security or privacy proof.

A boundary bug was found and fixed

When the exact selected count reaches zero, floating cancellation can leave a tiny residual in the stored moments. The decoder now returns the prescribed zero head while exposing that residue to the auditor. Both kernels also reject duplicate/retried, unknown and out-of-budget requests atomically without consuming additional horizon.

The memory risk is already visible

The following measurements use the same 100 Civil records, fixed 128-dimensional lexical curator, threshold 0.6 and deletion horizon eight. Only the lexical learner dimension changes. Each proposed state initially has 99 coefficient keys.



Learner dimension	B-E arrays (MB)	P-I arrays (MB)	P-I / B-E arrays
64	0.055	1.699	30.8x
128	0.175	6.641	37.9x
768	4.980	234.483	47.1x

These are initial allocated array bytes in decimal MB. They are not total physical memory, process peak, or serialized size. B-E includes eligible FP32 feature rows, FP64 targets and its current full Gram/cross matrices. P-I includes packed coefficient arrays. Metadata, matched public-ID adapters, allocator slack, external corpus/graph assets and workspaces are excluded from this table. Full physical accounting remains required.

The comparison is unfavorable to dense coefficient storage even though it already packs symmetric matrices. At 768 lexical dimensions, coefficient arrays use about 234 MB while baseline arrays use about 5 MB. This validates the need for the protocol's storage gate; it does not prove a universal lower bound or predict the results of the actual E5 model.

What changed during development

The initial harness varied curation dimension together with learner dimension. That confounded a dimension comparison. Its outputs were archived; the current pilot was rerun with a fixed curation graph and matched request paths. The amendment predates any confirmatory run and is recorded in ENGINEERING_CHANGELOG.txt.

Timing interpretation

Times are logged as unreplicated development observations. No speedup claim is made; full lifecycle costs, matched identifiers and hardware-blocked repetitions remain pending.

Acquisition and execution boundaries

Resource / gate	Observed status	Consequence
Local compute	9 visible CPUs; approximately 9.73 GiB reported physical memory	The proposed 128-GiB per-method study envelope is unavailable here. No GPU was detected.
Numerical stack	NumPy, SciPy and scikit-learn available	Exact-structure and convex-head engineering work runs here.
Transformer stack	No installed torch/transformers runtime or E5/MPNet weights	Primary semantic embeddings have not been computed.
Civil provenance	Current connector preview omits original article/publication/parent IDs	Article/source withdrawal is disabled for this slice; no source IDs are guessed.
News provenance	90 complete parsed rows; five observed hosts; two pages failed	This is not a complete source-group or representative corpus sample.
File fidelity	Untouched connector responses and derived-row hashes retained	Original corpus bytes and immutable Hub revisions remain unverified.
Human quality gate	Not performed	No semantic-duplicate precision or restored-information claim.

Completed software components

Exhaustive earlier-raw-neighbor graph construction; independently recomputed retained selection; source and record incidence rank; expected-admission formulas; cumulative R/S/U/A request generation; optimized eligible-row bookkeeping; indexed polynomial repair; common FP64 Cholesky decoding; memory preflight; checkpoint/resume; atomic failure checks; local ingestion validation; provenance manifests; and a failure-preserving result registry.

Next gate for the primary study

The source-rich original corpus assets and pinned model/runtime files must become available inside this workspace through a permitted import route. The current working connector exposes rendered previews rather than binary dataset/model downloads. The configured network allowlist does not provide a direct route to those hosts. No bypass or external training service was used.

Once those assets are available here, the next steps remain local: validate checksums and metadata, construct source-disjoint panels, compute pinned embeddings, complete human calibration, and run the native storage preflight before confirmation. A smaller actual memory cap must be locked and all infeasible cells retained. Synthetic replacements and silent substitution of lexical results for semantic results are excluded.

Reproducibility scope

The companion archive contains code, exact current outputs, archived development outputs, checksums, environment notes and the CC0 Civil engineering fixture. The raw CC-News text stays in the local working cache; the archive contains its provenance and audit results rather than redistributing the articles. All headline counts in this report resolve to saved JSON records. The main study manifest remains unlocked.